



Easy Futsal: Online Futsal Booking and Management System

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INTRODUCTION

Futsal is rapidly growing in Nepal, yet most booking centers still rely on phone calls, WhatsApp messages, and handwritten records leading to double bookings, payment disputes, and poor record-keeping. The Online Futsal Booking and Management System addresses this by providing a centralized web-based platform where users can search grounds, check real-time slot availability, book instantly, and pay securely through Khalti, while ground owners manage their listings and administrators oversee the entire platform through a single dashboard.

ACADEMIC QUESTIONS

How can a web-based booking system be structured to fully prevent double booking in futsal ground reservations while remaining secure, efficient, and easy to use for all stakeholders?

AIMS AND OBJECTIVES

- To develop a reliable web-based futsal booking system that reduces scheduling conflicts, prevents double bookings, and improves overall service efficiency.
- ❖ Implement real-time slot management to prevent double bookings.
- ❖ Ensure secure access and data protection for all users.
- ❖ Enable administrators to monitor bookings, payments, and generate reports efficiently.

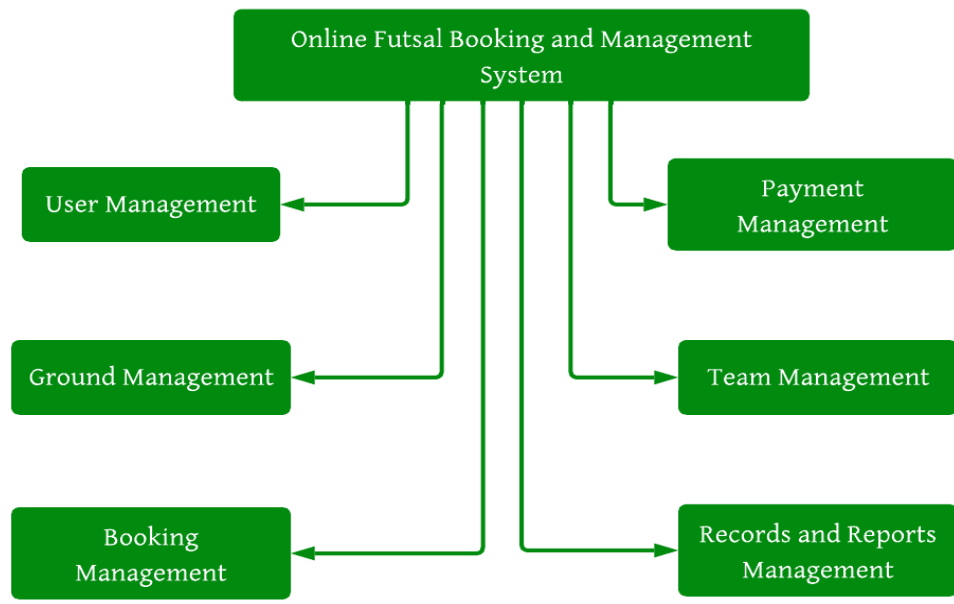
LITERATURE REVIEW

Research Papers	Key Findings
Web-Based Futsal Field Rent Information System in Bekasi	Improved booking accuracy, but lacked real-time updates, payments, and analytics.
First Come First Served Algorithm in Futsal Field Booking Applications	FCFS ensures fair scheduling, but missing real-time availability and payment features.
Futsal Booking Web-Based System Integrated with Telegram Notification	Telegram notifications speed up response and reduce admin workload.
Golden Goal Futsal Court Rental Mobile Application Using FCFS and Payment Gateway Integration	Payment gateway removes manual transactions, but lacks reminders and flexible scheduling.
Web-Based Sport Center Reservation System with Analytical Dashboard	Analytics dashboard improves decisions; needs better security and rescheduling features.

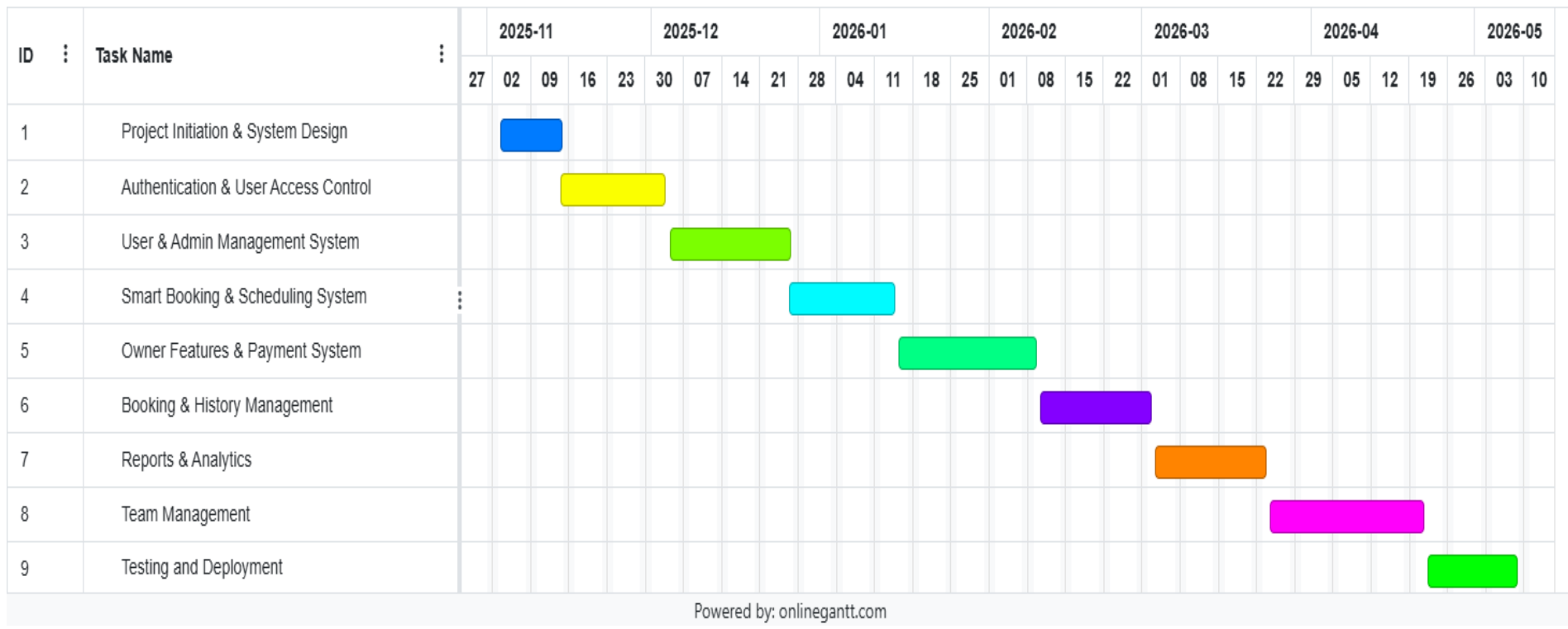
ARTEFACTS

Subsystem	Purpose / Key Functions
User Management	Handle user registration, login, and profiles
Ground Management	Manage futsal court details and availability
Booking Management	Handle reservations and schedule conflicts
Payment Management	Process payments, handle receipts, and invoices
Team Management	Manage team creation, membership, and invitations
Records & Reports Management	Generate reports and analytics for admin

FDD



GANTT CHART



TESTING

TestCase ID	Test Case	Result
TC01	User registration with valid data.	Pass
TC02	User registration with duplicate email.	Fail
TC03	Admin login with valid username and password.	Pass
TC04	Owner add new futsal ground with all details.	Pass
TC05	Owner update ground pricing and available facilities.	Pass
TC06	Admin accept owner registration request	Pass
TC07	User book a time slot with valid future date and time.	Pass
TC08	User attempt to book an already reserved time slot.	Fail
TC09	User view available slots via calendar interface.	Pass
TC10	Process online payment successfully via Khalti.	Pass
TC11	Khalti payment gateway service unavailable.	Fail
TC12	User view payment history with transaction status.	Pass
TC13	User create a team and invite players by username.	Pass
TC14	QR confirmation email sent after booking.	Pass
TC15	Admin generates monthly revenue report in pdf.	Pass
TC16	Team can be created and can challenge another	Pass

Evaluation/Reflection

- ❖ Successfully developed a functional and user-friendly futsal booking platform.
- ❖ Real-time slot management effectively reduced double booking problems.
- ❖ Khalti payment integration improved convenience and transaction security.
- ❖ QR-coded booking verification increased reliability and transparency.
- ❖ Understanding of full-stack web development and database management.
- ❖ Problem-solving abilities improved while handling challenges during system development.
- ❖ Overall system performance remains dependent on stable internet connectivity.

Conclusion

- ❖ The core problem of double booking was fully solved through real time slot blocking the moment a payment was confirmed.
- ❖ Users can search a ground, pick a time, pay online, and receive a QR code by email without any manual process involved.
- ❖ Ground owners got full control over their own listings, pricing, and booking records through a dedicated dashboard.
- ❖ Administrators can monitor everything happening across the platform from one central place.
- ❖ The Khalti payment gateway was successfully integrated and tested, making online transactions secure and straightforward for every user.
- ❖ A QR coded confirmation system was delivered successfully, giving users a reliable and verifiable proof of their booking at the ground.
- ❖ The team management feature gave players a way to create teams, send challenges, and track match history all within the same platform.

Future Scope

- ❖ Adding more payment options like eSewa or card payments so users are not limited to a single gateway when making a booking.
- ❖ Building a mobile app so people can search and book grounds directly from their phones without opening a browser.
- ❖ Introducing an AI powered recommendation engine that suggests the best grounds, time slots, and pricing based on a user's past booking behaviour and preferences.
- ❖ .Adding AI based smart pricing that adjusts ground rates automatically during peak hours and high demand periods, similar to how ride sharing apps work.
- ❖ Allowing team to team messaging so captains can coordinate match details, confirm challenges, and discuss schedules all inside the system.